Akash Pandey, Ph.D.

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EDUCATION

PhD, Machine Learning | Northwestern University | Evanston, IL, USA | *Sept 2021 - April 2026*
CGPA: 3.98/4.0, Advisors: Dr. Wei Chen and Dr. Sinan Keten

Masters, Applied Mechanics | Indian Institute of Technology Madras | Chennai, India | *Jul 2014 - June 2017*
CGPA: 9.4/10.0, Advisor: Dr. A Arockiarajan

TECHNICAL SKILLS

Programming Languages and Frameworks: Python, MATLAB, Pytorch, Numpy, Pandas, Jupyter Notebooks, Claude CLI, Captum, Scikit-Learn, AWS, Shell, Git, LAMMPS

Machine Learning Core: Transformer, Diffusion models, GAN, VAE, Explainable AI, LLM, LoRA, Random Forest, XGBoost, Recommendation Models, Active Learning, Gaussian Processes, Self-supervised learning, ProtBERT, ESM

Input Modalities: Univariate/multi-variate time series, banking and credit card data, protein sequence, RNA/DNA sequence, audio, biosignals, multivariate time series, multimodal, vision

Parallelism and Distributed Computing: DataParallel, PyTorch DDP (DistributedDataParallel)

SELECTED RESEARCH

Explainability-Steered Deep Generative Models for Biological Sequence Design *Jan 2026 - May 2026*
- Developed explainability-guided **VAE** and **diffusion** frameworks for protein/DNA design, achieving a 62% improvement in sample efficiency. (**under submission at NeurIPS'26**)

Explainability-Driven optimization under limited feedback for biological sequences *Aug 2025 - Jan 2026*
- Developed an **attribution-guided** evolutionary learning framework for **active learning**-based black-box optimization, improving **sample efficiency by 19%** across benchmarks. (**ICML'26, ICLR'26 MLGenX**)

Explainable (XAI) models for time series and biological sequences *April 2024 - July 2025*
- Designed novel explainable CNN-based deep learning architectures, `TimeSliver` and `COLOR`, improving explainability by **11%** on time-series data and by **22%** for biological sequences while maintaining Transformer-level predictability. **(ICLR'26, ICLR'25 MLGenX)**

Exploring electromyography (EMG)-to-Text Conversion with LLMs *Dec 2024 - Feb 2025*
- Devised an **LLaMA-3** based EMG-to-text model, reducing **word error rate** by 20% over baselines. (**ACL'25 Main**)

ACM Multimedia Challenge: Audio-based emotion detection *March 2023 - July 2023*
- Built an **emotion prediction model** using audio **foundation models** (**wav2vec** and **HuBERT**). Using **HuBERT** for audio embeddings improved performance by **4%** over **wav2vec**. (**ACM Multimedia'23**)

ICASSP Challenge: Person identification using biosignal from wearable device *Jan 2023 - March 2023*
- Architected a **multichannel CNN-based** deep learning model with a **late fusion technique** for person identification from **multivariate temporal biosignals**. The model boosted accuracy **from 62% to 91.36%**, earning **3rd place** in the challenge. (**ICASSP'23**)

WORK EXPERIENCE

Data Science PhD Intern | Capital One | San Jose, USA *Jun 2025 - Aug 2025*
- Implemented a **multi-GPU Transformer-based recommendation pipeline** to **rank ads** for the **Capital One mobile app**, delivering an **MVP** for production readiness and gained 20% **AUPRC** over the random forest baseline.
- Tested **10 XAI methods** (attention-, gradient-, perturbation-, and activation-based) to **quantify ad influence** in a transformer model; recognized **Attention×Gradient** as the top performer with a **10% improvement** over all other methods.

Data Science PhD Intern | Capital One | San Jose, USA *Jun 2024 - Aug 2024*
- To consider **users' history** for recommending ads, built an **ALBERT transformer**-based recommendation model to **rank ads** for the **Capital One webpage**, raising the **AUROC** score by **15%** over a random forest baseline.
- Leveraged **permutation-, Shapley-, and perturbation-based feature attribution methods** to evaluate input impact; the **ensemble method** identified the **top 5 of 40 features**, boosting interpretability and stakeholder trust.

Engineering Graduate to Engineering Lead | Rolls Royce & Infosys | Bangalore, India & Derby, UK *Jul 2017 - Aug 2021*
- Enabled **$5M in cost savings** by preventing unnecessary maintenance and ensuring timely component replacement through **linear regression models** that monitor aerospace engine health using flight data.
- Deployed a Siemens NX **UI feature** for simulation **model validation** and updates, enhancing model trust. Validation reduced model size by **10×**, cutting simulation time from **7 days to 1 day**.
- Formulated a **K-Means clustering** approach to segment material behavior into three regions, followed by fitting region-specific **nonlinear models**, improving R^2 by **30%**.